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Profile

Final-year Computer Science student targeting **quantitative research / research engineering**. I build end-to-end systematic-trading research—from raw tick ingestion and market-microstructure features, through an AFML-style ML validation pipeline, out to live execution—with a standing habit of **killing strategies that don't survive honest slippage, cost, and leakage testing**. Backed by production software-engineering experience in cloud services, async data infrastructure, and performance optimization.

Education

National University of Singapore — Singapore
Bachelor of Computing in Computer Science — *August 2023 – May 2027 (Expected)*

- NUS Merit Scholar; GPA: 4.58/5.00

Technical Skills

Quant / Research: market microstructure (VPIN, OFI, CVD, book imbalance, queue-position fills); AFML methodology (CUSUM bars, triple-barrier & meta-labeling, purged/embargoed CV & CPCV, sample weights, fractional differentiation, deflated Sharpe); slippage calibration; walk-forward backtesting
ML / Data: LightGBM, isotonic calibration, feature importance (MDA/SFI/permutation), pandas, numpy, PyTorch
Languages: Python, Go, C/C++, Java, TypeScript, SQL
Systems / Infra: numba JIT, PyArrow/Parquet, asyncio/WebSockets, HPC (SLURM), Docker, AWS (ECS/Fargate, S3, Lambda, CloudFormation), CI/CD, Linux, CUDA
Web: FastAPI, Flask, Express, React/Next.js

Independent Quantitative Research

Self-directed crypto systematic-trading stack, Feb–Jun 2026; every result below is grounded in a dated, reproducible artifact.

Systematic Research Lab — AFML ML pipeline on HPC | *Python, LightGBM, SLURM, numba* — May – Jun 2026

- Built an AFML pipeline (CUSUM event sampling, triple-barrier labels, purged/embargoed CV, sample weights, fractional differentiation) with a **non-negotiable lookahead auditor** gating every stage; deployed to a production config on an HPC **SLURM** cluster that held up under **16 bps** round-trip cost stress.
- Ran a streaming integrity audit over **3 years** of raw exchange data and found archive corruption—**33% non-monotonic** timestamps and **13M+ duplicate trades** inflating volatility features; patched the panels and published both the contaminated and corrected results.
- Transferred a tuned config to a new asset, caught that it had silently inherited the wrong (entry-only) slippage, and on rerun under correct round-trip friction **killed it** (11 of 15 walk-forward folds negative)—documenting the methodological error instead of the flattering first number.

Crypto Market-Making & Microstructure — backtest realism + systems | *Python, hftbacktest, PyArrow* — Feb – Apr 2026

- Found that an Avellaneda–Stoikov market-maker backtest was filling on transient price-touches: built a parallel fill model from order-book-delta-inferred trades and showed **~65% of fills were phantom**, then redesigned toward a static-quote model that cut fills **96%**.
- Fixed an unbounded memory leak in a **311-WebSocket** async collector—traced to PyArrow C++ column metadata living outside Python's allocator—cutting RSS drift from **2.4 GB to near-zero** over a 104-hour run.

ML-Gated Volatility Breakout — honest-cost kill + live bot | *Python, numba, Lighter/Binance* — Mar 2026 – Present

- Treated a backtest that looked too clean to be real as a red flag and audited it: found a self-inflicted ~10s lookahead leak, then an honest slippage model calibrated from **700+ real exchange fills** flipped the edge negative across **all 42 parameter cells**—killed the strategy class and shipped a reproducible negative-result package.
- Built and operate the **live execution bot**: routes DEX orders on Lighter off Binance signals with margin gating, latency profiling, and dollar-level fill reconciliation.

AFML Methodology Toolkit — learn-by-building | *Python, LLM agents* — Apr 2026

- To apply *Advances in Financial Machine Learning* rigorously, extracted 13 chapters into structured methodology files and built a **2,400-line agent orchestrator** that implements the pipeline with leakage gates baked into the tooling; reused three weeks later to boot the research lab above.

Software Engineering Experience

Software Engineer Intern — Jan 2026 – Jun 2026
Synapxe (Singapore's national HealthTech agency) — *Singapore*

- Designed and shipped **two internal monitoring dashboards** covering scheduled batch jobs and Oracle Service Bus integration traffic; deployed to staging on **AWS ECS Fargate** behind an internal ALB.
- Architected a reusable decoupled **producer/consumer pattern** (Python writer → **S3** → Node/Express + React SPA) provisioned via **CloudFormation**, cutting time-to-build for the follow-on dashboard.
- Authored 2,000+ lines across **React 19 / TypeScript**, **Express 5 / Python**, and CloudFormation IaC, reaching ∼**90%** test coverage.

Software Engineer Intern — Jun 2025 – Aug 2025
Cinch Asia — *Singapore*

- Launched two company-wide AI services in **Python** and **Docker**, reducing manual data processing by **30%**.
- Rewrote 8 legacy **AWS Lambda** scripts into consolidated **Golang microservices**, reducing cold-start latency by **75%** and cutting compute costs by **20%**.
- Implemented CI/CD and monitoring pipelines (**GitHub Actions**, Docker) to automate deployment and improve reliability.

Teaching Assistant | ***CS2100 Computer Organisation*** — Aug 2024 – May 2025
National University of Singapore — *Singapore*

- Tutored 100+ students weekly in data representation, pipelining, memory hierarchy, and assembly; built lab materials on processor architecture and performance optimization.

Activities

Quant Analyst (early exposure) — Sep 2023 – May 2024
NUS Investment Society — Quantitative Finance Department — *Singapore*

- First exposure to systematic trading: implemented a **mean–CVaR** portfolio optimizer and backtested allocation strategies on historical market data for the department's research track.

Selected Projects

BrainHack — TIL AI Competition | *Python, FastAPI, Docker, PyTorch, CUDA* — May – Jun 2024

- Co-led a 5-person team integrating **ASR**, **NLP**, and **VLM** models into an end-to-end multimodal pipeline served via a containerized **FastAPI** inference microservice (sub-100ms responses); **Finalist** among 50+ teams.